SPRING END SEMESTER EXAMINATION-2019

2nd Semester M.Tech / Ph.D Coursework

COMPUTATIONAL INTELLIGENCE

CS-6102

(For 2018 & 2017 Admitted Batches)

Time: 3 Hours | Full Marks: 50

Answer any SIX questions.
Question paper consists of four sections-A, B, C, D.
Section A is compulsory.
Attempt minimum one question each from Sections B, C, D.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) What is soft computing? How does it differ from hard computing? Differentiate between conventional AI and modern AI.

(b) Obtain 0.3-cut and strong 0.3-cut for following fuzzy set:

A = 0.2/-3 + 0.5/-2 + 0.2/-1 + 1/0 + 0.7/1 + 0.2/2 + 0.3/3 + 0.1/4

(c) Why sometimes Sigmoidal MF and LR MF are preferred over Gaussian MF and Bell MF?

(d) The MF of the fuzzy relation R = "x is much smaller than y" is defined as follows:

$\mu R(x,y) = \frac{y-x}{x+y+3}$, if $x < y$. and = 0, if $x \geq y$.

where $x \in X$ and $y \in Y$. If X = {2, 3, 4} and Y = {2, 3, 4, 5, 6}, then express the fuzzy relation R as a relation matrix.

as Algebraic Sum and complement operator as fuzzy classical complement operator, verify the above DeMorgan's laws for the following pair of fuzzy sets:

$A = 0.2/x1 + 0.6/x3 + 1/x4 + 0.7/x5 + 0.3/x6$

$B = 0.4/x1 + 0.7/x2 + 1/x4 + 0.5/x5 + 0.2/x6$

SECTION-C

4. (a) Explain the five defuzzification strategies which can be used in Mamdani Fuzzy model. Which of these according to you is the most preferable strategy? Justify. [4]

Explain diagrammatically fuzzy reasoning for three rules with two antecedents.

(b) Explain the steps of Genetic Algorithms (GA). Carry out at least two iterations of GA for the following optimization problem: [4]

Maximize $f(x) = 15x - x^2$ where $0 \leq x \leq 15$

(Assume probability of crossover as 100% and probability of mutation almost 0%)

5. (a) With some individual examples, explain the following terms associated with GA: [4]

Chromosome, Population (gene pool), Encoding scheme, Fitness evaluation, selection, crossover, mutation

(b) Write down eight activation functions (normally used in ANN) with necessary diagrams and formulations. [4]

6. (a) Explain a single neuron or single layer perceptron. Mention the perceptron learning rules. What is its main [4]

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6. (a) Explain a single neuron or single layer perceptron. Mention the perceptron learning rules. What is its main [4]

limitation? How is it handled through MLP in the context of XOR example?

(b) Develop an equivalent ANFIS architecture for a two-input first-order Sugeno fuzzy model with two rules. Draw the necessary diagrams. [4]

SECTION-D

7. (a) Develop a Mamdani fuzzy inference system diagrammatically with 2-rules 2-antecedents using min and max for T-norm and T-conorm operators respectively. Show the aggregated fuzzy output if Bell and/or Gaussian MFs are considered for antecedents and consequents. [4]

(b) Write down the characteristics of derivative free optimization. Elaborate on various types of stopping criteria. [4]

8. (a) Explain Generalized Modus Ponens (GMP). Why is this also called fuzzy reasoning or approximate reasoning? Draw the graphic interpretation of GMP using Mamdani's fuzzy implication and the max-min composition for following case: [4]

Multiple Rules with Multiple Antecedents

(b) "MLP training is possible due to Back propagation Learning Rule"- Explain the significance of this statement with some analytical view point. [4]

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